

Question: A 3-month-old infant is brought to her pediatrician because she coughs and seems to have difficulty breathing while feeding. In addition, she seems to have less energy compared to other babies and appears listless throughout the day. She was born by cesarean section to a G1P1 woman with no prior medical history and had a normal APGAR score at birth. Her parents say that she has never been observed to turn blue. Physical exam reveals a high-pitched holosystolic murmur that is best heard at the lower left sternal border. The most likely cause of this patient's symptoms is associated with which of the following abnormalities?

Options: (A) 22q11 deletion (B) Deletion of genes on chromosome 7 (C) Lithium exposure in utero (D) Retinoic acid exposure in utero

Domain Experts

Question domains:

Pediatrics

Cardiology

Pulmonology

Neonatology

Option domains:

Cardiology

Genetics

Question Analyses

...It's important to manage VSD promptly to prevent complications such as congestive heart failure, pulmonary hypertension, and growth failure.

...VSD is a congenital heart defect, meaning it is present at birth, and it is not related to the mode of delivery or the APGAR score.

...Cyanosis is often seen in infants with significant left-to-right shunting of blood, but in this scenario, the absence of cyanosis suggests that the VSD is small to moderate in size.

...Small VSDs may close spontaneously over time, while larger VSDs may require surgical intervention to prevent complications.

Option Analyses

Option A: The symptoms...are consistent with a VSD

Option B: ...a deletion of genes on chromosome 7

Option C:...

Option D:...

Option A: ...

Option B: ...

Option C: ...not known to cause ventricular septal defects....

Option D: ... be associated with a range of birth defects

Initial Report

Key Knowledge:

 Clinical assessment of an infant with symptoms suggesting VSD...

Total Analysis:

 The infant's symptoms are consistent with VSD... Options such as 22q11 deletion, deletion of genes on chromosome 7, lithium exposure in utero are not relevant to the given scenario.

	✓		✓		✓		✓		✗	,the report should...
	✓		✓		✓		✓		✗	,the report should...
	✓		✓		✓		✓		✓	

Unanimous Report

Key Knowledge:

 The infant's symptoms are concerning for a possible congenital heart defect or a respiratory condition...

Total Analysis:

 ...one of the most common genetic abnormalities associated with congenital heart defects, including VSD, is the 22q11 deletion syndrome, also known as DiGeorge syndrome...